

## **Colombia**

### **Superintendence of Industry and Commerce**

As reported in 2021, the Superintendence of Industry and Commerce (SIC) has been developing three projects aimed at improving technological tools and analytical techniques that can help the SIC's investigators and forensic engineers to meet the challenges posed by the large volumes of data produced and consumed by today's markets. In addition, in 2022, the SIC has begun to develop a new data analytics tool named 'Búho.'

These projects were envisioned as tools that could: enhance the Authority's detection capabilities; increase its investigative efficiency when addressing large volumes of structured and unstructured data; pinpoint suspicious signs, patterns, transactions, and exhibit pricing insights within such data, and better allocate its human resources.

The current results can be summarized as follows.

#### **I. Sabueso**

Sabueso is a price monitoring data tool aimed at enhancing the authority's inspection, surveillance, and control functions. It has the objective of harnessing available data on products offered by supermarkets on their websites. This tool allows a baseline of information to analyze the historical behavior of products that have been or that are or could be under the scrutiny of the SIC.

During 2021 and 2022, Sabueso captured information on one (1) store in different categories. For 2023, the coverage will be increased to five (5) market stores, and the tool will be programmed: (i) to detect possible anticompetitive conducts between the monitored agents, and (ii) to send alerts directly to the authority.

#### **II. Inspector**

This data analytics tool developed by the SIC supports the monitoring process by the Competition Advocacy Group of different regulatory projects published on the websites of state regulators. This tool alerts the authority of published regulatory projects detected on their official websites that may impact competition.

The tool has a model for identifying regulation projects that could affect the competition dynamics in the market. It performs natural language processing (NLP) and converts the document into a vector that will go through a classification model to determine the possible impact of the project on competition matters. The model has been trained from the history of concepts issued by the Competition Advocacy Group.

In July 2022, Inspector was monitoring and reporting relevant information of 38 out of 68 different Colombian authorities. In comparison, in April 2023 Inspector was monitoring 60 out of the 68. This means, Inspector increased its regulatory projects issued capability detection from 55.8% to 87.1% the monitored authorities.

In conclusion, Inspector has evolved in a highlighted way due to its effectiveness in gathering information on new regulatory projects issued by regulatory entities to analyze later whether such projects may impact free competition.

### **III. Sherlock**

Sherlock is the data analytics tool project that seeks to support the SIC's investigators in the identification of signs or patterns that suggest potential anticompetitive behaviors on public procurement processes (bid-rigging).

This data analytic tool is currently in its second stage. This means that the Data Analytics Group is working on automation of the search of the abovementioned signs and patterns within the bulk of public procurement data available online based on machine learning technologies and deep machine learning. It relies on artificial intelligence (AI).

This tool hasn't been used to identify possible bid-rigging conducts because it is still in a developing stage.

### **IV. Búho**

This data analytics tool is meant to support the SIC in the detection of different news or reports circulating online, on different websites, that could be potentially related to anticompetitive behaviors. This way, the authority gets the alert and can evaluate the scope of the reported facts.

Búho is still in an early developing stage. Currently, the Data Analytics Group of the SIC is identifying: (i) which websites are going to be the source of information; (ii) key words to generate the alert; and (iii) the best way to warrant the effectiveness of the notification to each related group.